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Large Language Models and Responsible AI in Clinical Decision Support Systems: A Systematic Literature Review

Dennis Paulino[a], André T.C. Netto[b], Rafael Ris-Ala[a], Artur Rocha[a] and Hugo Paredes[b]

[a]HumanISE, INESC TEC, Porto, Portugal; [b]University of Trás-os-Montes e Alto Douro, Vila Real, Portugal

ABSTRACT
The incorporation of large language models (LLMs) into clinical decision support systems (CDSS) offers potential for enhancing the quality of healthcare. Nevertheless, there is a dearth of comprehensive standards and understanding about the application of explainable and responsible AI practices in this context. The objective of this study is to present an overview of the literature on the utilization of LLMs in CDSSs, with an emphasis on Explainable AI (XAI) techniques and responsible AI principles. The present study was conducted through a systematic literature review analyzed of 36 studies on the application of LLMs in CDSS, with a focus on XAI and responsible AI. Human-centered AI is essential to design CDSSs that not only provide accurate recommendations but also align with clinicians' trust and workflow realities. The results suggest that while LLM-based CDSSs demonstrate potential in enhancing clinical decision-making, concerns regarding model interpretation into healthcare workflows persist as challenges.

KEYWORDS
Clinical decision support systems; responsible AI; trustworthy AI; large language models; human–computer interaction (HCI)

1. Introduction

Decision Support Systems (DSS) are computer-based tools that assist organizations in making informed decisions by analyzing large volumes of data (Alasiri & Salameh, 2020; Broby, 2022; Dastres & Soori, 2021; Mumali, 2022; Rahman et al., 2021), with significant applications in healthcare (Mahiddin et al., 2022; Spoladore & Pessot, 2021). Extract-Transform-Load (ETL) processes are indispensable for converting unstructured data into structured formats suitable for analysis. These processes are also of great importance for harmonizing raw data sources into standardized formats, which can enhance the accuracy of DSS. For example, Nan et al. (2022) state the increasingly necessity for harmonizing disparate healthcare data sources, including medical imaging, genomics, and clinical records. This is a necessary step to eliminate biases and variability across datasets, which is crucial for enhancing the reproducibility and reliability of Artificial Intelligence (AI) models in healthcare and ensuring consistent clinical insights across diverse settings. In healthcare, it is also recommended that research focuses on the further development of harmonization techniques with a view to enhancing the capacity of AI to operate effectively with diverse, large-scale healthcare datasets and to deliver more accurate predictions (Nan et al., 2022). In parallel with these developments, recent AI-driven Clinical Decision Support Systems (CDSS) have demonstrated significant advances in medical image segmentation and computer-aided diagnosis through deep learning and transformer-based architectures applied to complex conditions such as gliomas, liver cancer, and bone tumors (Lu et al., 2025; Ye et al., 2025; Zhang et al., 2026). The advancement of AI models, particularly those classified as generative AI, such as large language models (LLM), has had a considerable impact on the field of CDSS, specifically in the process of biomedical text mining. Transformer-based language models such as BioBERT and ClinicalBERT, which have been

CONTACT Dennis Paulino dennis.l.paulino@inesctec.pt HumanISE, INESC TEC, Porto, Portugal
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specifically designed to handle the nuances of health-related language, have effectively addressed the shortcomings of general natural language processing (NLP) models in processing specialized medical terminology (Huang et al., 2019; Lee et al., 2020). Precise data transformation is essential to handle the inherent complexities of biomedical language, facilitating the development of biomedical pre-trained language models like PubMedBERT for tasks such as information extraction and text classification in healthcare (Kalyan et al., 2022). Addressing these challenges highlights the necessity of domain-specific pre-training and meticulous ETL processes for LLMs to excel in specialized fields like biomedicine. LLMs like GPT-4 are designed to handle natural language processing tasks, including data extraction and transformation. The incorporation of LLMs into ETL workflows has markedly advanced the automation and scalability of medical data processing (Ullah et al., 2024). The incorporation of LLM technologies, including GPT-3.5, LLaMA 2, and Vicuna 7B, within the ETL pipeline has facilitated the automation of essential patient data extraction from unstructured medical reports, thereby eliminating the necessity for manual searches (Ullah et al., 2024). In this context, it is important to distinguish between generative LLMs, such as GPT-based architectures, and transformer-based encoder models, such as BERT and its variants. While both belong to the transformer family, only generative models are considered LLMs in the strict sense of this study. Encoder-based models are therefore treated as supporting NLP components within CDSS pipelines rather than primary LLM systems.

Notwithstanding these developments, significant obstacles remain in the complete exploitation of LLMs for the extraction of medical information. Studies have investigated the zero-shot capabilities of models such as ChatGPT and GPT-4 in extracting oncologic phenotypes from unstructured radiological reports without prior retraining (Fink et al., 2023; Hu et al., 2024). These studies underscore the importance of human supervision when deploying LLMs in medical settings to ensure data accuracy and reliability. The growing demand for explainability and transparency in AI systems is particularly evident in the context of complex models such as deep neural networks, as evidenced by the case of LLMs. The lack of interpretability inherent to such models represents a significant obstacle to their adoption in critical sectors such as healthcare, law, and finance (Barredo Arrieta et al., 2020). In response to this issue, the European Union's proposed AI Act[2] introduces a risk-based regulatory framework that categorizes AI applications into distinct levels of risk, including high-risk, limited-risk, and minimal-risk. This framework aims to ensure appropriate levels of oversight and emphasizes transparency and human intervention for high-risk systems (Veale & Zuiderveen Borgesius, 2021). The ALTAI assessment list from the EU (HLEG of the EU Commission, 2020) delineates seven fundamental tenets for the development of reliable and trustworthy AI systems. These systems must be designed to enhance human agency and oversight, while ensuring the implementation of robust human oversight mechanisms, such as the "human-in-the-loop" approach. Furthermore, the development of AI systems must prioritize the protection of user privacy and the security of user data. It is of the utmost importance that AI systems are governed in accordance with data protection laws, such as the General Data Protection Regulation (GDPR) (Ryz & Grest, 2016). This necessitates the implementation of practices such as data minimization, encryption, and privacy by design, which are essential for maintaining user trust. These developments are in alignment with the objectives of XAI, which include the promotion of fairness, accountability, and privacy, as well as the assurance that AI models are not only accurate but also ethical and trustworthy in real-world applications (Barredo Arrieta et al., 2020). XAI is important in the ETL process when utilizing LLMs, given that these models can be opaque, rendering decisions without discernible rationale (Busra et al., 2024). To address this issue, XAI techniques such as SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) can be employed to enhance comprehension of model predictions. SHAP is a game-theory-based method that assigns an importance value to each feature in a model's prediction, thereby ensuring that the contributions are consistent and additive across all features (Lundberg & Lee, 2017). LIME provides an explanation of the prediction of any machine learning model by approximating it with a simpler, interpretable model in the local vicinity of a particular prediction (Ribeiro et al., 2016).

The concept of human-centered AI (HCAI) is often subject to ambiguity due to the variety of definitions that emerge across domains such as human–computer interaction (HCI) and AI ethics. These domains are concerned with the design of AI systems that prioritize human values and agency. A study identifies four principal areas of research in the field of human-centered AI: explainable and interpretable AI, human-centered design methods, human-AI teaming, and ethical AI (Capel & Brereton, 2023). These areas are primarily concerned with the integration of human-in-the-loop approaches, which entail the incorporation of human oversight and interaction to enhance AI decision-making in a manner that aligns with human values and contextual needs. In the context of healthcare, the incorporation of human-in-the-loop methodologies is of paramount importance for the adaptation of LLMs to clinical settings. This enables the optimization of these models by facilitating the validation and interpretation of AI-generated insights by medical professionals. The handling of sensitive data within ETL processes necessitates a robust emphasis on data privacy and security. The implementation of responsible AI practices ensures compliance with established ethical standards, thereby promoting fairness, transparency, and data privacy (Veale & Zuiderveen Borgesius, 2021). It is imperative to conduct ongoing assessments of LLMs to prevent the propagation of biases in extracted or transformed data, thereby ensuring the integrity of the data and maintaining stakeholder trust.

Several systematic literature reviews have identified the growing significance of XAI in the context of CDSSs, emphasizing the need for AI models that are both accurate and interpretable in healthcare settings (Kim et al., 2024). In one systematic review, studies employing XAI techniques in healthcare AI models were examined, and three principal categories of healthcare datasets were identified: clinical features, text, and high-dimensional data (Loh et al., 2022). The findings underscore the necessity for reliable AI models to not only demonstrate optimal performance but also to yield interpretable outcomes. Antoniadi et al. (2021) proposed in a systematic literature review that trust can be established through XAI techniques that facilitate transparency in decision-making processes, thereby reducing the opacity of AI models and ensuring responsible deployment. Furthermore, the importance of user-centric AI design is underscored, with techniques such as visualizations and HCI methods enhancing accessibility and interpretability, which are fundamental to responsible AI (Xu et al., 2023). An interdisciplinary approach integrating insights from technology, law, medicine, and ethics is proposed to ensure compliance with legal and ethical standards, particularly regarding GDPR and healthcare regulations. It is anticipated that LLMs will have a transformative impact on medicine, enhancing diagnostic precision and aiding clinical decision-making (Karabacak & Margetis, 2023). However, successful integration necessitates the resolution of challenges pertaining to transfer learning, domain-specific fine-tuning, expert input, dynamic training, interdisciplinary collaboration, appropriate evaluation metrics, clinical validation, ethical considerations, data privacy, and regulatory compliance. By adopting a multifaceted approach and fostering collaboration, LLMs can be developed and integrated into medical practice in a responsible manner, thereby enhancing patient care and outcomes. Another review indicates that ChatGPT and analogous AI tools have the potential to significantly alter the landscape of radiology reporting. By transforming unstructured medical data into structured templates, these tools have the capacity to enhance accuracy and standardization, thereby improving the quality of radiology reports (Sacoransky et al., 2024). The current limitations include concerns regarding data privacy, the potential for medical errors, and a lack of specialized medical training. In the absence of tailored prompt engineering, there is a risk of incorrect or incomplete information being generated. It is recommended that techniques such as dynamic few-shot prompting and retrieval-augmented generation (RAG) be integrated to enhance the effectiveness of clinical workflows.

In view of the aforementioned considerations, the present study aims to conduct a systematic review of the literature on the application of LLMs in clinical decision support systems. The advent of generative AI and its potential to enhance DSS in various domains, including specially in healthcare systems, underscores the necessity to comprehend the opportunities and challenges intrinsic to AI systems. The objective is to examine how explainable AI, responsible AI practices, and human-in-the-loop approaches can enhance CDSSs with the advent of LLMs, while ensuring ethical compliance. The contributions of our systematic review are two-fold:

We provide the first systematic literature review of the recent literature on the integration of LLMs into clinical decision support systems and the adoption of responsible AI principles within the healthcare context.

We analyze key challenges and trends, highlighting persistent gaps related to model generalizability, interpretability, and clinical workflow integration, as well as regulatory, ethical, and governance barriers to safe and effective deployment of human-centered AI-driven CDSSs.

2. Methodology

2.1. Research questions

This work was conducted in accordance with the guidelines proposed by Kitchenham (2004) for conducting and reporting literature reviews in the domain of software engineering. In general terms, this method allowed for the evaluation, aggregation, and synthesis of evidence from the available literature, taking into account its relevance to understanding a phenomenon, topic area, or research problem using a systematic and rigorous approach. Prior to formulating the research questions (RQs) for a systematic literature review, Kitchenham (2004) recommends first defining a set of structural elements embedded into the PICO (Population, Intervention, Comparison, and Outcomes) strategy:

- Population – Clinical Decision Support Systems
- Intervention – Generative AI toward Responsible AI;
- Comparison – Other Clinical Decision Support Systems that doesn't focus on Generative AI;
- Outcomes – Accuracy, fulfillment of some of the Responsible AI or Trustworthy AI principles on Clinical Decision Support Systems;

    - The six principles for Responsible AI from Barredo Arrieta et al. (2020): Responsible Fairness, Privacy, Accountability, Ethics, Transparency, Security & Safety.
    - The seven principles for Trustworthy AI from HLEG of the EU Commission (2020): Human Agency and Oversight; Technical Robustness and Safety; Privacy and Data Governance; Transparency; Diversity, Nondiscrimination, and Fairness; Societal and Environmental Well-being; Accountability.

In light of the growing emphasis on developing solutions that enhance CDSS with generative AI, this SLR encompasses an eight-year research period (2017–2024) on CDSSs that integrate generative AI with the principles of Responsible or Trustworthy AI. This range was selected based on the fast pace of generative AI such as the development of the first generative pre-trained transformer (GPT), known as GPT-1[3], in 2018, which may cause the older solutions of generative AI to be less effective and sometimes outdated. Based on the PICO strategy, the RQs that guide this article to accomplish the goal of this SLR are the following:

- RQ1. What are the current main solutions on CDSSs with AI?
- RQ2. What research topics are being addressed in the intersection between Generative AI, Responsible AI/Trustworthy AI and CDSSs?
- RQ3. What are the limitations of current research, and how should we approach them?

2.2. Search process

The search process is focused on primary studies from search engines (i.e. Scopus) and digital libraries (i.e. ACM Digital Library, IEEE Xplore, PubMed). Most of these electronic sources have been suggested as relevant in the field of software engineering (Brereton et al., 2007). In the pursuit of records in the health domain, PubMed (including PubMed Central) was used with the purpose of gathering insights on clinical studies. The search string was then constructed taking into consideration the defined PICO structure and the following steps:

1. Analyze the RQs to identify search terms;

Table 1. Search string used in the primary search phase.

Scope	String
Large language models	("Large Language Model" OR "Language Model" OR "BERT" OR "GPT" OR "ChatGPT")
Clinical decision support system used in healthcare context	("Clinical Decision Support" OR CDSS OR (("Decision Support" OR DSS) AND (Health OR Healthcare))
Extract-transform-load process for handling of data	("ETL" OR "Extract Transform Load" OR "unstructured data")
Principles toward a responsible or trustworthy AI	(XAI OR "explainable AI" OR "local explanation technique" OR "model interpretability" OR "Responsible AI" OR "AI Ethics" OR Fairness OR Privacy OR Accountability OR Ethics OR Transparency OR Security OR Safety OR "Human Agency" OR Oversight OR "Technical Robustness" OR Privacy and "Data Governance" OR "Diversity" OR "Non-discrimination" OR "Societal Well-being" OR "Environmental Well-being")

2. Identify search terms in relevant articles;
3. Identification of synonyms and alternative spellings for the search terms;
4. Construction of search strings using Boolean Operators (i.e. AND and OR).

At a higher level, the search string presented in Table 1 was adapted to each electronic source and formulated in accordance with its main scope. In addition to model-related terminology, the search strategy deliberately incorporated terms related to ETL processes and unstructured data handling. This decision was conceptually aligned with the Responsible and Trustworthy AI dimensions defined in the PICO framework, particularly those concerning privacy, data governance, technical robustness, and security. In clinical environments, generative AI-based CDSSs do not operate in isolation at the model level; rather, they are embedded within complex data pipelines that extract, transform, and integrate sensitive patient information from heterogeneous sources such as Electronic Health Records, laboratory systems, and clinical documentation. The handling of such data through ETL mechanisms directly affects compliance with data protection regulations, transparency of data provenance, and the reliability of downstream AI-driven recommendations. Therefore, the inclusion of ETL-related terms was intended to capture studies addressing not only generative model capabilities but also the infrastructure-level processes necessary for responsible deployment in real-world healthcare settings.

2.3. Inclusion criteria

The inclusion criteria (IC) comprehend the RQs defined previously and involve studies published in or after 2017 and before 31 December 2024. This date range has the purpose of characterizing the current state-of-the-art approaches for generative AI in CDSSs with the implementation of Responsible AI principles. The inclusion criteria followed in this study are presented as follows:

1. Articles that include generative AI applied on CDSSs having in consideration some of the Responsible AI principles;
2. Studies published between 2017 and 31 December 2024.

2.4. Exclusion criteria

The exclusion criteria (EC) aim to screen the studies collected and then, remove the duplicates along with studies not published in English and that were not peer-reviewed. In order to achieve this, the following exclusion criteria were defined:

1. Duplicate reports of the same study (when several reports appeared, it was only accepted the most completed version);
2. Not written in English;
3. Full article not available;
4. Editorials, keynotes, abstracts, tutorials, dissertations or theses;
5. Not peer-reviewed;

6. Not related to the topics addressed in the research questions;
7. Studies without an empirical validation or experimental results; and
8. Studies published before January 1, 2017 and after 31 December 2024.

2.5. Data collection

The data collected from each article was:

- Source of Publication (Conference, Journal), Year and Authors;
- Research Question/Purpose of the study and Technologies used (i.e. tools used to construct the solution);
- Evaluation methods, Target population and Main findings;
- Summary of the article and Additional Notes (Complementary observations on the study);

Given the heterogeneity of the included studies in terms of design, data sources, and evaluation approaches, a lightweight structured quality assessment was conducted to support the interpretation of the evidence. Rather than applying a single standardized risk-of-bias tool across diverse study types, we adopted a pragmatic classification based on key dimensions relevant to clinical AI systems. Each study was evaluated according to (i) study design (e.g. retrospective, simulated, prospective), (ii) data realism (synthetic or simulated data versus real clinical data), and (iii) validation context (technical performance evaluation, comparison with clinical benchmarks, or clinician-facing assessment). Based on these criteria, studies were categorized into three levels of evidence: exploratory, corresponding to proof-of-concept studies relying on simulated scenarios, indirect evaluation, or synthetic data; moderate, corresponding to retrospective analyses using real clinical data without integration into clinical workflows; and higher-maturity, corresponding to studies involving clinician interaction, comparison with expert decision-making, or evaluation in near real-world or workflow-integrated settings.

2.5.1. Conducting the review

The study selection process was conducted by two reviewers (the principal author and a co-author) to ensure methodological rigor and reduce selection bias. In the initial phase, titles and abstracts were screened independently according to the predefined inclusion and exclusion criteria. Discrepancies between reviewers were discussed and resolved through consensus, ensuring a consistent interpretation of the criteria. Full-text assessment was conducted based on the same criteria, with uncertainties resolved through discussion when necessary, leading to the final inclusion of 36 studies. Figure 1 presents the flow diagram of the search process. A total of 1,207 records were identified through database searching. After the application of predefined inclusion and exclusion criteria, 1,106 records were retained for title and abstract screening, of which 1,028 were excluded. A total of 78 articles were assessed for full-text eligibility, and 42 were excluded based on relevance and lack of empirical validation. This resulted in a final set of 36 studies included in the review. Additionally, the flowchart of the article selection process presents a view of the records excluded, taking into account the exclusion criteria defined in Section 2.4. The results were then analyzed and synthesized in order to answer the RQs formulated in Section 2.1.

3. Results

On a general level, the results allow us to characterize the research conducted on LLMs in health systems through a critical lens, in accordance with the principles of Responsible and Trustworthy AI. Based on the defined quality assessment framework, the included studies were distributed across three levels of evidence: exploratory ($n = 6$), moderate ($n = 23$), and higher-maturity ($n = 7$). A detailed mapping of studies to evidence levels is provided in the supplementary material. Exploratory studies typically relied on simulated scenarios or proof-of-concept evaluations, while moderate studies predominantly used retrospective analyses of real clinical datasets. Higher-maturity studies involved clinician-facing evaluation, comparison with expert performance, or near real-world validation settings.

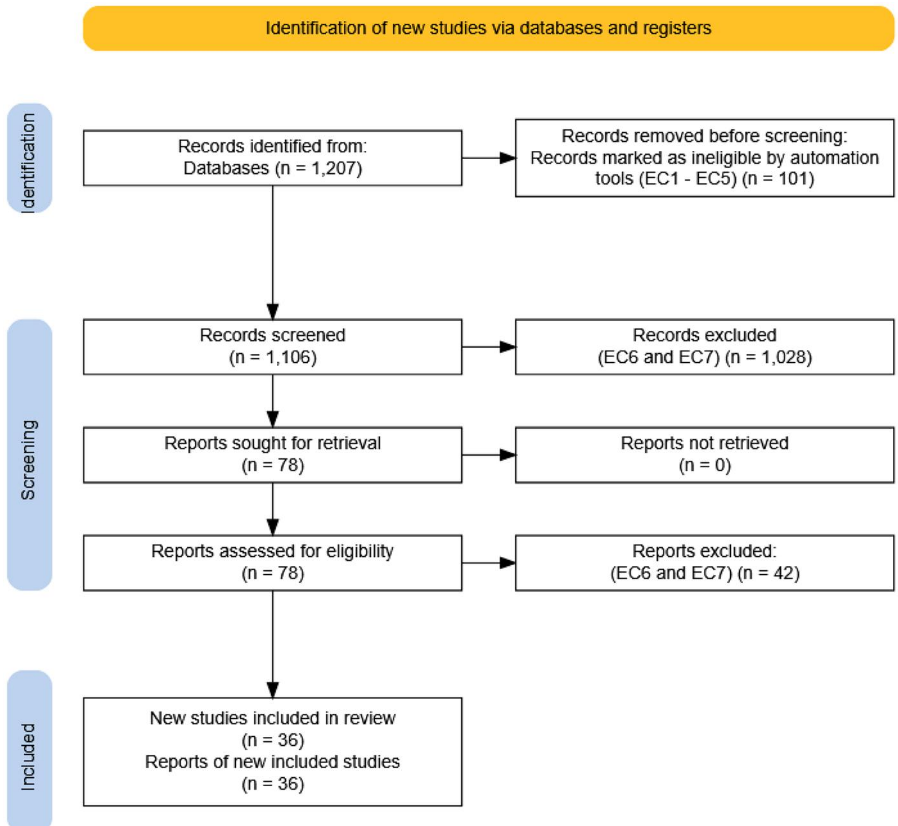

Figure 1. Flowchart of the study selection process (PRISMA-style representation (Haddaway et al., 2022)).

As shown in Figure 2, there has been a substantial growth in the number of publications on this topic in the last four years. The most popular publication venues for research on this topic are Diagnostics (MDPI) (8.3%), Quantitative Imaging in Medicine and Surgery (AME Publishing) (5.6%), Scientific Reports (Springer Nature) (5.6%), and the Conference on Human Factors in Computing Systems (CHI) (5.6%). Notably, the majority of studies were published in journals and conferences with a primary focus on health and medicine or health informatics, including venues such as JAMA Network Open, Journal of Multidisciplinary Healthcare, and JMIR Medical Informatics. This distribution highlights the interdisciplinary nature of the field, with collaborations spanning health sciences, informatics, and computer science.

To assess the impact of the inclusion of ETL-related terms, we examined the contribution of these terms to the final study selection. A total of 4 out of the 36 included studies were identified through references to unstructured data, corresponding to the ETL-related component of the search string. This indicates that the inclusion of ETL-related terms did not substantially influence the overall scope of the review, but rather complemented the identification of infrastructure-level aspects relevant to responsible AI deployment. The association of each included study with ETL-related terms was documented in the supplementary material.

3.1. Technologies

The following section presents a more detailed examination of each of the principal topics identified in the preceding synthesis. Each section will provide a comprehensive elaboration of the underlying concepts, trends, and implications for clinical practice and research, drawing from the diverse range of articles and approaches presented in the dataset (see Table 2 for an overview of the technologies employed in the selected articles).

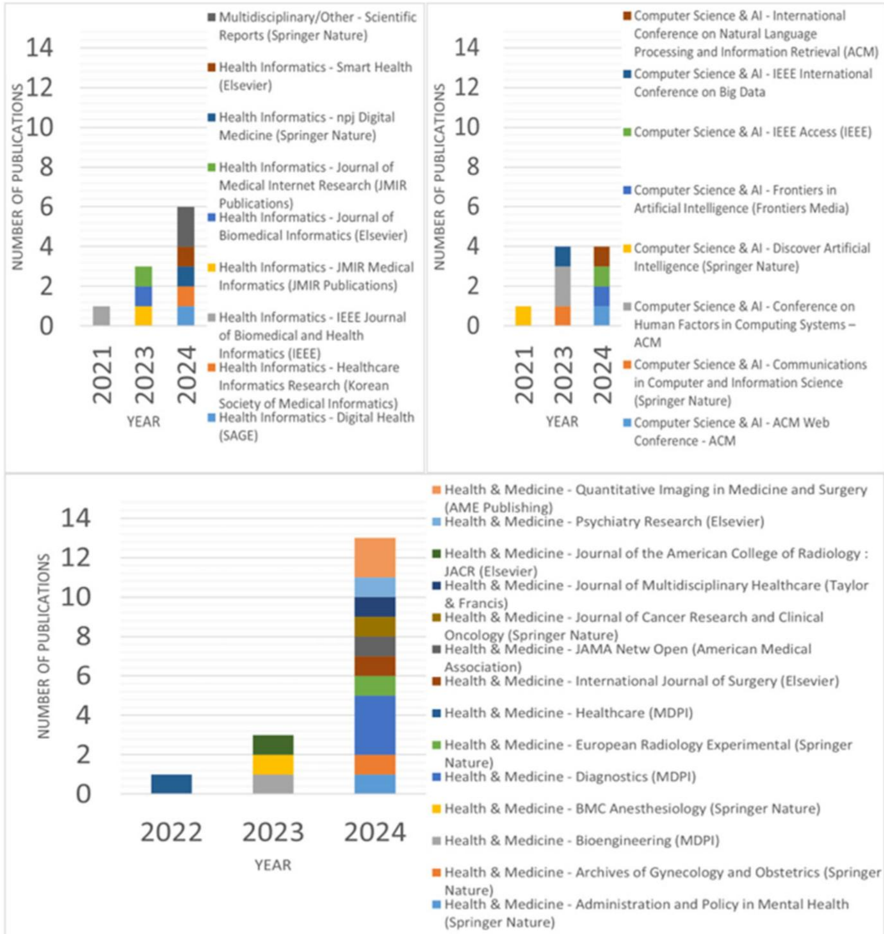

Figure 2. Number of publications in various journals and conference venues over the years. To support the thematic grouping of journals and conference venues, the Scopus database (https://www.scopus.com/) was used as a reference for subject area classification: Health Informatics and multidisciplinary/other (top left), computer science & AI (top right) and health & medicine (bottom).

3.1.1. Generative LLMs and prompt engineering

Generative large language models (LLMs), such as GPT-3.5 and GPT-4, have been increasingly applied to tasks including clinical reasoning, diagnostic decision support, and healthcare workflow optimization (Arnaud et al., 2023; Wada et al., 2024). For instance, GPT-3.5 and GPT-4 have demonstrated effectiveness in providing radiological decision support (Rao et al., 2023), assisting with emergency department triage (Shah-Mohammadi & Finkelstein, 2024), and supporting neuroradiology diagnostics through role-playing and step-by-step reasoning prompts (Wada et al., 2024). These models derive significant benefit from prompt engineering techniques, such as chain-of-thought reasoning and scenario-specific prompts, which encourage more structured and interpretable outputs. Systems integrating GPT-4 have also demonstrated efficacy in telemedicine-based management of metastatic prostate cancer and diagnostic reasoning in clinical evaluation settings.

3.1.2. Transformer-based models (BERT variants, clinical BERT, BioBERT) for NLP

Transformer-based models such as BERT and its specialized variants (BioBERT, Clinical BERT, FlauBERT, CamemBERT, and MCN-BERT) are increasingly employed for the processing of clinical text, leveraging domain adaptation to enhance the comprehension of medical terminology. These BERT-like systems are not considered LLMs in the generative sense and are treated as transformer-based encoder models used for representation and classification tasks. A number of studies have demonstrated the versatility of these models in a variety of tasks, ranging from admission prediction to symptom classification. Some efforts have relied on CamemBERT or FlauBERT for forecasting

Table 2. Overview of the technologies used in the selected studies.

Article	Language model type	PE	CM	DL	NLP	XAI	DI	At
Ahmed et al. (2024)	LLM (GPT), CNN	✓	–	CNN	–	Interpretability	Health guidelines	–
Rao et al. (2023)	LLM (GPT-3.5/4)	✓	–	–	–	–	Clinical scenario	–
Wada et al. (2024)	LLM (GPT-4)	✓	–	–	–	–	Radiology cases	–
Lalk et al. (2024)	Encoder (BERT)	–	–	–	BERT-topics	XAI for topics	–	–
Arnaud et al. (2023)	Encoder (CamemBERT, FlauBERT)	–	–	–	French text prep	LIME	ED context	–
Dal et al. (2024)	LLM (ChatGPT-4)	–	–	–	–	Concordance check	Oncology telemedicine	–
Goh et al. (2024)	LLM (GPT-4)	–	–	–	–	Basic performance metrics	Comparison with clinical tools	–
Hassan et al. (2024)	Encoder (MCN-BERT)	–	–	–	–	Prec/Rec/F1 only	–	–
Diniz et al. (2022)	Encoder (BERT-based)	–	–	–	Text preprocessing	–	Mental health monitoring	–
Alloatti et al. (2021)	Encoder (BERT-based NLU)	–	–	–	NLP parsing	–	Diabetes guidelines	–
Nazary et al. (2024)	LLM (ChatGPT – GPT-3.5)	✓	RF, LR, XGB guide LLM	–	KNN Imputation	SHAP via ML	Domain knowledge gen.	–
Shah-Mohammadi and Finkelstein (2024)	LLM (GPT-3.5, GPT-4)	✓	–	–	–	Stats accuracy	ICD-10 mapping	–
Naseem and Catal (2024)	Encoder (BioBERT)	–	–	–	Data augment.	Attention weights	MIMIC-III domain	–
Yang et al. (2024)	Encoder (BERT) + TabNet	–	–	–	–	SHAP	Rehab outcomes	Fairness
Zhang and Ogasawara (2023)	Encoder (BERT embeddings)	–	NB, SVM baselines	[Res/Alex]Net, VGG11, Bi-LSTM	Word2Vec	Grad-CAM for text	–	–
Chen et al. (2024)	LLM (ChatGLM)	–	–	LSTM+MLP	–	–	HIV/TB domain	–
Gao et al. (2023)	Encoder (BERT, ClinicalBERT)	–	–	–	ICD-10 Ontology	–	Yes (A-P relations)	–
Pham et al. (2024)	LLM (ChatGPT-4)	–	–	–	–	Manual error analys.	Nephrology triage	–
Li et al. (2024)	LLM (ChatGPT-4) vs KB	–	–	–	NLP Q & A	–	Prostate surgery KB	–
Md Hasib et al. (2024)	Transformer (DCNN main)	–	ML benchmarks	DCNN	–	LIME, SHAP	Pediatric SLI	–
Rajashekar et al. (2024)	LLM + ML (LASSO, RF, GBT)	–	LASSO, RF, GBT (Gradient Boosted Trees)	DL also	Embeddings	Partial dependency plots	GI bleeding CDSS	Retrieval augment.
Oh et al. (2024)	LLM (GPT-3.5, GPT-4)	–	GBM benchmark	–	–	Accuracy metrics	KSA sepsis DB	–
Chung et al. (2023)	Encoder (BioClinicalBERT)	–	RF, SVM	–	fastText, TF-IDF	SHAP	ASA-PS	–
Sun et al. (2024)	LLM (ChatGPT-3.5)	✓	–	–	–	CoT interpretability	BI-RADS ultrasound	–
Yang et al. (2023)	LLM (GPT-3)	–	–	–	Literature retrieval	User acceptance	Evidence-based DST	–
Griewing et al. (2024)	SLM vs LLM (GPT-3.5, GPT-4)	–	–	–	–	Concordance metrics	German breast cancer guideline	Local hosting & security
Griewing et al. (2024)	LLM (GPT-3.5, GPT-4, LLaMA2, Bard)	✓	–	–	–	Concordance with MTB	German breast cancer guideline	–
Puts et al. (2023)	Encoder (FlauBERT) + NLP (fastText, Word2Vec)	–	GBT	–	TF-IDF, embeddings	SHAP	Coding complexity	MLOps deployment
Rau et al. (2024)	LLM (GPT-4) + retrieval	–	–	–	text-embedding/ada	Stats (Mann-Whitney)	GI imaging KB	Hyperlinking sources
Zhang and Song (2024)	LLM (GPT-2)	–	–	–	Tokenization/Encoding	CUQ usability test	Chronic disease diag.	–
Savage et al. (2024)	LLM (GPT-3.5, GPT-4)	✓	–	–	–	McNemar's, Kappa	MedQA, NEJM cases	CoT reasoning (XAI)
Fei et al. (2024)	LLM (GPT-3.5, GPT-4)	–	–	–	–	ICC for reliability	Cognitive testing	–
Meng et al. (2021)	Transformer (BRLTM)	–	LSTM, GRU baselines	–	–	Attention visualization	Depression prediction	–
Rao et al. (2023)	LLM (GPT-3.5, GPT-4)	–	–	–	–	Accuracy, recall	Clinical workflow	–
Liu et al. (2024)	Transformer	–	LR, RF, GB as baseline	–	–	SHAP	Nutritional	–
Wang et al. (2024)	LLM (GPT-4)	✓	–	–	–	TI-RADS accuracy	ThyroAIGuide	–

Legend: Language Model Type: Indicates the type of language model used, distinguishing between generative LLMs (e.g. GPT-based architectures) and transformer-based encoder models (e.g. BERT, BioBERT, ClinicalBERT), which are not considered LLMs in the strict generative sense; PE (Prompt Eng.): Prompt engineering techniques; CM (Classical ML): Traditional ML (e.g. LR, RF, XGBoost, SVM); DL (Non-LLM): Deep learning without LLMs (CNN, LSTM, etc.); NLP (Non-LLM): Traditional NLP feature extraction (TF-IDF, Word2Vec, etc.); XAI: Explainable AI tools (SHAP, LIME, Grad-CAM, CoT reasoning for interpretability); DI (Domain Integration): Use of clinical guidelines, ontologies, knowledge bases; AT (Additional Tech): Blockchain, fairness optimization, design fiction, retrieval augmentation, etc. For models or techniques, we provide short references (e.g. "RF" for Random Forest, "XG" for XGBoost, etc.).

emergency department admissions (Arnaud et al., 2023) and on MCN-BERT for medical concept normalization (Hassan et al., 2024).

For instance, BioBERT has been employed to prioritize readmissions in clinical data (Naseem & Catal, 2024) and, in combination with TabNet, to handle multimodal data for psychiatric rehabilitation (Yang et al., 2024). Meanwhile, BERT embeddings have been used to analyze psychotherapy transcripts via BERTopic (Lalk et al., 2024) and to facilitate conversation management in diabetes care (Alloatti et al., 2021). In certain instances, BERT variants are integrated with simpler downstream models or additional architectures to achieve optimal results. For instance, BioClinicalBERT has been employed for tasks such as ASA-PS classification (Chung et al., 2023), while Clinical BERT has been utilized for assessment-plan relation extraction in progress notes (Gao et al., 2023). Researchers who do not utilize BERT's generative capabilities often benefit from its context-aware embeddings, as demonstrated by a BERT-based model for suicidal ideation classification in Brazilian Portuguese (Diniz et al., 2022) and by studies extracting embeddings for Convolutional Neural Network (CNN)s, BiLSTMs, or Random Forest (RF) classifiers (Zhang & Ogasawara, 2023). In more specialized environments, models such as MCN-BERT adapt BERT's lexical knowledge to handle specific medical domains, thereby reducing error from domain shift (Hassan et al., 2024).

A significant benefit of these models is their domain-specific pretraining, which is frequently conducted on biomedical corpora. Through learning from extensive medical text, BERT variants develop proficiency in recognizing complex healthcare terminology and relationships, enhancing tasks such as named entity recognition and concept normalization (e.g. ICD-10 coding). This domain-focused approach serves to mitigate the inaccuracies that can emerge when general-purpose LLMs are employed in specialized fields. Rather than relying solely on BERT for classification or sequence labeling, many studies incorporate its embedding into traditional machine learning classifiers or recurrent neural architectures. This hybrid method preserves BERT's semantic insights while reaping the computational efficiency or interpretability benefits of simpler models, such as random forests or bidirectional long short-term memory networks. This approach is particularly appealing in clinical contexts where interpretability and resource constraints are paramount. BERT variants have proven valuable for a range of clinical document analyses, including ED admission forecasts, readmission risk assessments, and patient complexity evaluations. For instance, Meng et al. (2021) introduced a Transformer-based BRLTM (Bidirectional Representation Learning with Transformer Model) pretrained on EHR data for depression prediction, underscoring how domain adaptation can significantly enhance clinical outcomes. The incorporation of ontologies or databases, such as SNOMED or ICD-10, into these models serves to refine their contextual understanding, thereby enabling more precise insights into patient conditions and trajectories.

3.1.3. NLP techniques & preprocessing

NLP techniques have been shown an important role in text classification, sentiment extraction, and coding complexity prediction, even as LLMs and BERT variants gain traction. A number of studies have explored the use of classical approaches such as TF-IDF (Term Frequency–Inverse Document Frequency), one-hot encoding, and Bag-of-Words, as well as word embedding methods like Word2Vec and fastText, to transform unstructured clinical text into features for downstream predictive tasks. For instance, investigators have adopted word embeddings (Word2Vec, fastText) and text transformations for medical text classification (Zhang & Ogasawara, 2023) or used BioClinicalBERT and fastText embeddings for tasks like ASA-PS classification (Chung et al., 2023). NLP-based feature engineering extends beyond simple classification. In some cases, chain-of-thought prompting (an advanced NLP technique) enhances interpretability and accuracy in large language model use cases (Savage et al., 2024). These efforts are underpinned by fundamental steps such as tokenization, lemmatization, and normalization, which are essential for handling domain-specific vocabulary and out-of-vocabulary words in clinical text data. The classical feature extraction methods (e.g. TF-IDF, Bag-of-Words) frequently function as baseline approaches for the representation of unstructured text, such as clinician notes or radiology reports. Although they provide limited context in comparison to BERT embeddings, they remain computationally efficient and transparent for end users—a particular consideration where model explainability is paramount. Word2Vec and fastText embeddings capture semantic relationships between medical terms, adapting effectively to multilingual or domain-specific data, as evidenced by their use in generating robust features for tasks such as suicide risk assessment or medical coding complexity. In French-language contexts, FlauBERT is employed, demonstrating the adaptability of embedding methods to varying linguistic needs (Arnaud et al., 2023). These processes underscore the versatility of NLP pipelines, which persist as the foundation of numerous clinical decision-making frameworks, particularly where specialized text preprocessing and interpretability remain paramount.

3.1.4. XAI and interpretability tools

XAI methods, including SHAP, LIME, Grad-CAM, partial dependency plots, and chain-of-thought reasoning, have been extensively adopted to enhance clinician confidence and transparency in AI-driven systems. In the context of emergency department triage predictions, LIME provides local interpretability by illuminating how input features influence (Arnaud et al., 2023). Furthermore, the combination of SHAP with traditional feature importance analysis has been demonstrated to guide advanced language models such as ChatGPT (Nazary et al., 2024). SHAP has also been employed in time-series or multimodal setups, elucidating how a BERT-TabNet model integrates textual and structured data for psychiatric rehabilitation (Yang et al., 2024).

In the context of text classification, Grad-CAM has been demonstrated to highlight the critical words that influence predictions (Zhang & Ogasawara, 2023), while LIME and SHAP have been shown to elucidate the results of DCNN for specific language impairment detection (Md Hasib et al., 2024). For instance, SHAP has been employed to ascertain the nutritional factors most influential in determining Alzheimer's disease mortality (Liu et al., 2024), reflecting the applicability of XAI to diverse healthcare use cases.

In the field of medical imaging and pathology, methods such as SHAP offer a unique approach to explainable AI by providing both global explanations, which consider the feature importance across an entire dataset, and local explanations, which focus on patient-specific inputs and alterations in outcomes. LIME (Local Interpretable Model Agnostic), another notable method, employs a similar strategy by perturbing inputs and observing changes in outputs. These dual-level insights are invaluable for validating models on both aggregated and individual bases. Grad-CAM (Gradient-based Camera Calibrated Networks) and attention heatmaps provide a visualization of the relevant image or text regions, which is crucial in radiology and pathology contexts. By verifying that the model's focus corresponds to recognized pathology (e.g. tumor regions or abnormal clusters), clinicians can gain confidence in the AI's decision-making process and identify possible errors if the highlighted area appears to be of little clinical relevance. Tools such as LIME and SHAP are largely model-agnostic, meaning they can be applied to random forests, gradient-boosted trees/GBT (XGBoost), CNNs, or large language models alike. Conversely, chain-of-thought prompting is more LLM-specific in that it encourages models to articulate their intermediate reasoning steps. Irrespective of the approach adopted, the overarching goal is to build clinician trust, guide model refinement, and ensure accountability – essential qualities in healthcare, where the cost of an opaque or mistrusted prediction can be high.

3.1.5. CDSSs, domain knowledge integration, and clinical guidelines

A considerable body of research highlights the significance of integrating knowledge bases, clinical guidelines, and ontology-based reasoning into AI models, thereby ensuring adherence to established standards of care. For instance, specialized prompt engineering and multimodal AI approaches address healthcare-specific requirements (Ahmed et al., 2024), while conversational agents adhere to evidence-based protocols for diabetes management (Alloatti et al., 2021). To anchor AI-driven recommendations more firmly in clinical reality, researchers embed domain knowledge in dedicated knowledge bases. For instance, robotic-assisted prostate surgeries benefit from the RARPKB repository, which combines a knowledge base with LLM comparison (Li et al., 2024). A similar hybrid approach is underscored by decision-support systems that merge LLM interfaces with ML risk dashboards (Rajashekar et al., 2024). The incorporation of gradient boosting models (GBMs) with LLMs in sepsis mortality prediction serves to illustrate how knowledge-driven AI has the capacity to elevate the status of existing analytic tools

(Oh et al., 2024). The integration of clinical guidelines as explicit frameworks serves to ensure that the outputs of AI remain relevant to everyday practice. For instance, the integration of breast cancer guidelines into a small language model (SLM) and LLM frameworks provides comprehensive decision support in oncology (Griewing et al., 2024). Adherence to standardized protocols, such as the BI-RADS or the DSM-5, serves to mitigate the occurrence of inconsistent or arbitrary recommendations. ThyroAIGuide offers a practical illustration by uniting GPT-4 with domain rules for thyroid nodule assessments (Wang et al., 2024).

Systems that encode clinical guidelines – whether for German breast cancer care or robotic-assisted prostatectomy – ground model outputs in recognized expert consensus. This approach aligns AI recommendations with accepted treatment pathways, thereby increasing clinician confidence and reducing unwarranted variability in patient care. Many decision-support frameworks fuze multiple methods. Projects such as GutGPT (Rajashekar et al., 2024) combine natural language question-answering from LLMs with ML-based risk prediction dashboards, delivering user-friendly interactions while maintaining robust, guideline-informed analytics. The result is a flexible decision-support environment that clinicians can navigate via plain language queries. Clinicians require evidence to justify or reject AI-driven recommendations. The integration of summaries of relevant biomedical literature, as demonstrated in a trust-calibration framework (Yang et al., 2023), enables physicians to adopt AI outputs with confidence. The support of a recommendation with references to reputable studies reduces the perceived "black box" effect and fosters acceptance. Consequently, knowledge-based systems integrated with AI not only enhance diagnostic or therapeutic accuracy but also uphold transparency and maintain fidelity to clinical standards.

3.2. Evaluation methods

The articles employ a diverse array of evaluation strategies to assess the performance, reliability, interpretability, and clinical applicability of AI and ML models in healthcare (see Table 3). The evaluations encompass a range of methodologies, from traditional accuracy and AUC metrics to user-centered usability testing, qualitative interviews, error analyses, and sophisticated interpretability tools such as SHAP and LIME. Comparisons are frequently made against baselines, expert benchmarks, clinical guidelines, and other language model versions. The aforementioned evaluation methods are collectively designed to guarantee the trustworthiness, clinical relevance, fairness, and seamless integration of models into healthcare workflows. The evaluation frameworks presented in these articles are diverse and multifaceted, combining a range of quantitative metrics (accuracy, AUC, F1) with interpretability assessments (LIME, SHAP, Grad-CAM), user acceptability studies, fairness checks, and domain-specific guideline adherence tests.

3.2.1. Performance metrics and baseline comparisons

A considerable number of studies in this field employ well-established metrics, including accuracy, precision, recall, F1-score, and AUC-ROC, to evaluate the performance of the models. For instance, Goh et al. (2024) emphasize the importance of diagnostic accuracy and efficiency. To provide a contextual framework for new methods, studies often employ baseline comparisons against classical machine learning or simpler architectures. Other works, such as Nazary et al. (2024), benchmark ChatGPT's performance relative to interpretable ML models, thereby illustrating whether more sophisticated large language models justify their added complexity. Transformer-based techniques also follow this tradition. Naseem and Catal (2024), for example, shows how domain-adapted approaches like BioBERT fare compared to traditional ML or simpler neural architectures. The ultimate objective of these evaluations is to ascertain whether advanced solutions, such as large language models, domain-specialized transformers, or novel hybrid systems, deliver meaningful gains over established methods.

Error analysis is also frequently implemented as a method of identifying common failure cases. For instance, Wada et al. (2024) analyze misdiagnoses in neuroradiology, while Fei et al. (2024) examine false positives and negatives to refine models. Cross-validation (e.g. Chung et al., 2023) and ablation tests (e.g. Li et al., 2024) are employed by several studies to ensure the generalizability and stability of models.

Table 3. Overview of the evaluation methods and target population of each selected study.

Article	Performance metrics	Model/method comparisons	Explainability	Target population
Ahmed et al. (2024)	Accuracy, Precision, Recall, F1-score	Compared PE framework vs. baselines	Qualitative interpretability	Healthcare datasets (e.g. X-rays)
Rao et al. (2023)	Accuracy against guidelines, reasoning quality	ChatGPT-3.5 vs. GPT-4, OE vs. SATA prompts	Error analysis of reasoning, interpretability checked	Simulated clinical cases (breast cancer screening)
Wada et al. (2024)	Diagnostic accuracy, error rates	Effect of PE techniques on GPT-4 Turbo	Qualitative analysis of explanations/reason clarity	Neuroradiology cases from simulated diagnostic scenarios
Lalk et al. (2024)	Topic coherence, accuracy, precision, recall, F1	ML models predicting therapy outcomes (BERTopic)	XAI methods to link topics and outcomes	Psychotherapy transcripts (indirectly patients in therapy)
Arnaud et al. (2023)	Accuracy, Precision, Recall, F1, AUC-ROC	Compared CamemBERT vs. FlauBERT models	LIME for interpretability of key words/phrases	ED patients (triage notes, predicting admissions)
Dal et al. (2024)	Concordance w/ oncologist decisions, accuracy, efficiency	ChatGPT 4.0 compared with medical oncologists	Qualitative insights from oncologists	Cases of prostate cancer in telemedicine scenarios
Goh et al. (2024)	Diagnostic accuracy, efficiency	Physicians w/ChatGPT vs. traditional	LLM performance	Physicians (diagnostic reasoning)
Hassan et al. (2024)	Precision, Recall, F1, Accuracy	MCN-BERT vs. other transformer models	Error analysis	Clinical data of patients
Diniz et al. (2022)	Precision, Recall, F1, AUC-ROC	BERT-based model performance monitored	Qualitative feedback from mental health professionals	Individuals at risk of suicidal ideation (user texts)
Alloatti et al. (2021)	F1 score BERT model, usage metrics	Manual validation by a linguistics expert	Qualitative design workshops	People with diabetes, caregivers
Nazary et al. (2024)	Accuracy, Precision, Recall, F1, cost-sensitive	Compared ChatGPT vs. RF, LR, XGBoost, baseline models	SHAP for interpretability ML model guiding ChatGPT	Heart Disease dataset, multiple hospitals
Shah-Mohammadi and Finkelstein (2024)	Match rates at category & body system levels	GPT-3.5 vs. GPT-4 differential diagnoses	ICD-10-CM coding & CCSR mapping (interpretability)	ED patients (MIMIC-III), 24-h of admission for diagnosis
Naseem and Catal (2024)	AUROC, F1-score, valid. accuracy	BioBERT vs. other transformers/ML methods	Attention mechanism usage	Patients (MIMIC-III) risk re-admission
Yang et al. (2024)	Accuracy, Precision, Recall, F1, AUROC	Multimodal (BERT+TabNet) vs. baseline models	SHAP for explainability, fairness checks	Patients with mental illness in psychiatric rehabilitation
Zhang and Ogasawara (2023)	Accuracy, Precision, Recall, F1, AUROC	[Res/Alex]Net,VGG11 vs. NB,SVM, Bi-LSTM, Word2Vec vs. BERT	Grad-CAM for text explanations	Patients spanning multiple medical domains (from medical texts)
Chen et al., 2024)	Accuracy, Precision, Recall, F1, AUROC, AUPRC	LSTM+MLP vs. baseline ML models	No explicit XAI mentioned	HIV patients at risk of TB (structured EMR data)
Gao et al. (2023)	F1, Precision, Recall, Accuracy	Hybrid model w/ICD-10 vs. vanilla BERT	Error analysis	Patients represented by clinical notes
Pham et al. (2024)	Accuracy, inter-rater agreement	Benchmark vs. clinician gold standard	Manual error analysis	Simulated nephrology patient text
Li et al. (2024)	Accuracy, Relevance (expert-review)	RARPKBvs.ChatGPT-4 (recommendations)	Expert review of recommend.	Patients robot-assisted prostatectomy
Md Hasib et al. (2024)	Accuracy, Precision, F1, AUROC	DCNN vs. LR, Decision Trees, Transformers	LIME, SHAP	Pediatric Patients suspected of SLI
Rajashekar et al. (2024)	No standard metrics stated	LLM + ML (LASSO, RF, GBT) integration vs. baseline ML/CDSS	Interpretability	EM & IM residents, med students (teams interacting with a CDSS)
Oh et al. (2024)	Accuracy, Precision, F1, AUROC	ChatGPT-4, GPT-3.5 vs. GBM benchmark	–	Sepsis patients (database) predicting mortality
Chung et al. (2023)	Accuracy, Precision, F1	BioClinicalBERT, fastText, RF, SVM	SHAP	Preoperative patients (anesthesia context)
Sun et al. (2024)	Accuracy, Cohen's Kappa for agreement	ChatGPT-3.5 vs. human radiologists of varying expertise	CoT reasoning for interpretability	Patients undergoing breast ultrasound; evaluating GPT-3.5's BI-RADS classification
(Yang et al. (2023)	Accuracy of trust calibration (accept correct AI suggestions)	Literature-evidence-based DST vs. traditional explainability methods	Qualitative feedback on UI, usability, trust	Clinicians verify AI suggestions with biomedical literature
Griewing et al. (2024)	Concordance rates	SLM vs. GPT-3.5/4.0 vs. tumor board	Transparency & explainability	Breast cancer patients indirectly; SLM simulating guideline recommend.

(*continued*)

Table 3. Continued.

Article	Performance metrics	Model/method comparisons	Explainability	Target population
Griewing et al. (2024)	Agreement percentages, accuracy across complex cases	GPT-3.5, GPT-4, Llama2, Bard compared to MTB	Error analysis for evolving LLM outputs	Simulated high-complexity breast cancer patient profiles
Puts et al. (2023)	F1, Accuracy, confusion matrix, agreement ratio	Compared expert coder agreement, majority class baseline, GBT model	SHAP for interpretability (feature importance)	Medical coding processes in hospitals
Rau et al. (2024)	Accuracy, Consistency	GPT-4 retrieval-augm. Chabot vs. generic g-CB	Statistical tests (Mann-Whitney), reasonable checks	Patients needing GI radiology differential diagnosis (GPT-4 retrieval-augm. chatbot)
Zhang and Song (2024)	Accuracy, AUC, Precision, Recall, F1	Fine-tuned GPT-2 model no explicit baseline mention	CUQ usability questionnaire for user experience	Patients with chronic diseases seeking preliminary diagnostic support
Savage et al. (2024)	Accuracy, McNemar's Test, Cohen's Kappa	GPT-3.5 vs. GPT-4, CoT prompts vs. other diagnostic reasoning prompts	Logical rationale analysis by physicians	Physicians as end-users (diagnostic reasoning scenarios)
Fei et al. (2024)	ICC for reliability, Accuracy	ChatGPT vs. traditional clinician-led cognitive evaluations	Iterative refinement of scoring algorithms	Healthy individuals and stroke survivors for cognitive assessment tasks
Meng et al. (2021)	AUC-ROC, Precision, Recall, F1, Accuracy	BRLTM vs. LSTM, GRU, logistic regression, other baselines	Attention visualization for interpretability	Patients at risk of depression (EHR data)
Rao et al. (2023)	Accuracy, recall, precision	ChatGPT vs. clinician responses or standards	Systematic usability study	Diverse patient conditions (simulated)
Liu et al. (2024)	AUC-ROC, Precision, Recall, F1, Accuracy	Transformers vs. logistic regression, RF, GB	SHAP for nutritional factor explanation	Individuals with Alzheimer's disease
Wang et al. (2024)	Accuracy, Cohen's Kappa, sensitivity, specificity	GPT-4 vs. human experts with various experience levels	Chain-of-thought reasoning to enhance interpretability	Patients with thyroid nodules (ultrasound data), clinicians

3.2.2. Explainability and user centered evaluation

A significant number of articles integrate XAI techniques, including LIME, SHAP, and Grad-CAM, with the aim of elucidating the influence of specific features or image regions on predictions, thereby fostering trust among clinicians. For instance, Arnaud et al. (2023) and Nazary et al. (2024) employ LIME and SHAP to reveal key factors in model outputs, while Zhang and Ogasawara (2023) rely on Grad-CAM for visually interpretable imaging tasks. In addition to quantitative metrics, several studies employ qualitative methods. For instance, trust calibration is another common theme, and it was explored by Yang et al. (2023) in how referencing reputable biomedical literature aids clinicians in deciding whether to accept or reject AI-generated recommendations. Research on user-centered evaluation frequently underscores clinical workflow efficiency and physician acceptance. Dal et al. (2024) assesses time savings and usability in telemedicine environments, while Rajashekar et al. (2024) emphasize user trust in AI outputs. Chung et al. (2023) and Liu et al. (2024) strengthen these conclusions by testing AI solutions across a range of clinical datasets. This confirmed the versatility and practicality of such systems in day-to-day healthcare settings.

3.2.3. Domain-specific validation and guideline concordance

Many studies underscore the importance of aligning AI-driven outputs with recognized clinical standards and expert judgements, thereby ensuring their genuine clinical relevance. To this end, Rao et al. (2023) and Shah-Mohammadi and Finkelstein (2024) benchmark their models' decisions against well-established radiology and emergency department guidelines (e.g. the American College of Radiology Appropriateness Criteria), confirming the methods' concordance with real-world practice. Li et al. (2024) validated a prostate cancer surgery decision-support system founded on a dedicated knowledge base by comparing its recommendations against those of expert oncologists and contemporary AI models. This insistence on evaluating performance not just statistically, but also via expert feedback and guidelines, guarantees that the findings remain both robust and clinically meaningful.

3.3. Target population

The target populations across these studies are diverse, encompassing patients with various conditions, healthcare professionals (e.g. radiologists, pathologists, clinicians), and on occasion, simulated or indirect patient scenarios (see Table 3). A significant number of studies employ de-identified, actual patient data derived from electronic health records (EHRs), imaging, or clinical notes. In other cases, studies rely on synthetic, standardized, or historical datasets representing specific patient cohorts. These may include patients at risk of heart disease, pediatric patients with leukemia, individuals with mental health conditions, or patients with complex conditions such as metastatic prostate cancer or thyroid nodules. It is also noteworthy that several studies concentrate on the requirements and workflows of healthcare providers (including physicians, mental health professionals and radiologists), reflecting a dual focus: the indirect improvement of patient care through the provision of AI-driven decision support tools for clinicians.

Whilst a significant number of AI-driven healthcare solutions are implemented at a single site, several studies have highlighted the broader geographical applicability and linguistic adaptations of such solutions. Arnaud et al. (2023) employs French-language triage data, and Alloatti et al. (2021) designs conversational agents for Italian-speaking users. These examples highlight the importance of tailoring AI models to specific linguistic and cultural contexts, ensuring that solutions remain effective and equitable across diverse healthcare landscapes.

3.3.1. Direct patient cohorts from clinical settings and specific populations with special needs

A considerable body of research focuses on real-world patient data associated with specific clinical conditions. For instance, certain studies address chronic or complex diseases, such as HIV patients at risk of tuberculosis (Chen et al., 2024). Additionally, research targeting depression risk draws on large-scale electronic health record data (Meng et al., 2021). Imaging-based investigations examine cohorts undergoing specialized tests, such as chest X-rays or thyroid ultrasounds (Wang et al., 2024).

A subset of research targets vulnerable groups or individuals with unique clinical requirements. Pediatric populations frequently serve as the focal point of research endeavors. For instance, Md Hasib et al. (2024) attend to children aged 0–6 suspected of having specific language impairment. In the context of mental health, Yang et al. (2024) and Meng et al. (2021) focus on individuals grappling with serious mental illnesses or at risk of depression, underscoring the importance of domain-focused solutions for these at-risk cohorts. Further research has been conducted on stroke survivors and patients with chronic diseases. For instance, Fei et al. (2024) conducted a study to evaluate cognitive assessments among stroke survivors and healthy adults. In contrast, Zhang and Song (2024) explored the potential of AI to assist in diagnostics for patients managing chronic illnesses. Collectively, these studies demonstrate how tailored AI interventions can better serve populations requiring specialized attention, thereby advancing precision in healthcare delivery.

3.3.2. Indirect or simulated patient scenarios

Some studies opt for standardized or synthetic patient cases rather than real patient data, allowing them to explore AI models in controlled, reproducible conditions. For instance, Rao et al. (2023) and Dal et al. (2024) utilize simulated or fictional clinical vignettes to assess large language models (GPT-3.5/4) in radiology decision-making and telemedicine oncology management, respectively, thereby ensuring that the systems can manage realistic clinical complexities without compromising patient privacy. Wada et al. (2024) employs neuroradiology cases from the AJNR to evaluate GPT-4 Turbo, simulating diagnostic workflows without direct involvement of patient data. By leveraging these structured yet representative scenarios, researchers can systematically evaluate AI-driven solutions while minimizing risks associated with real-world implementations. Simulated breast imaging scenarios have been shown to help guide radiological decision-making (Rao et al., 2023). These clinical populations commonly originate from substantial datasets, such as MIMIC-III (Johnson et al., 2016) or hospital-specific EHR systems, which capture a wide spectrum of patient demographics and disease severities.

3.3.3. Healthcare providers as a target user group

A number of studies emphasize the role of healthcare professionals, including clinicians, radiologists and mental health specialists, as the key end-users of AI systems. For instance, Lalk et al. (2024) utilize psychotherapy transcripts to indirectly assist therapists and facilitate routine outcome monitoring. In a similar vein, Diniz et al. (2022) introduce the "Boamente" tool, enabling mental health professionals to monitor suicidal ideation in real time. By engaging these targeted user groups, researchers ensure that AI-based interventions seamlessly integrate into existing clinical workflows, ultimately enhancing patient care rather than disrupting established practice norms.

3.4. Responsible AI

Most articles place an emphasis on interpretability and transparency as the fundamental principles of responsible artificial intelligence, reflecting the clinical necessity for intelligible AI outputs in high-stakes healthcare settings (see Table 4). The most addressed principle was Transparency ($n = 31$ studies) and the least addressed were Fairness ($n = 3$ studies) and Accountability ($n = 3$ studies). Furthermore, the issues of privacy and ethics are frequently addressed, with a considerable number of studies utilizing de-identified data and aligning their methodologies with established clinical guidelines and patient safety objectives. The concept of accountability is frequently manifested through the presence of human oversight. It is common practice to present AI systems as decision-support tools, rather than autonomous agents. The issue of fairness is less frequently addressed directly, although some studies do mention fairness optimization or consider diverse patient populations. Security and safety are frequently implied by rigorous evaluation, local hosting, or blockchain frameworks that guarantee data integrity. The articles demonstrated a collective awareness that responsible AI in healthcare must be responsible, interpretable, and ethically grounded.

3.4.1. Transparency as a core focus

Transparency and interpretability are frequently cited as fundamental principles in numerous publications, with the objective being to ensure that clinicians can comprehend and repose their trust in AI-driven outcomes. Techniques such as SHAP, LIME, Grad-CAM, and chain-of-thought prompting provide insights into the internal logic of models, allowing healthcare providers to verify their accuracy and clinical alignment. For instance, Arnaud et al. (2023) utilize LIME to emphasize the most influential words in emergency department admission predictions, thereby assisting clinicians in validating the model's reasoning. In a similar vein, Nazary et al. (2024) have similarly incorporated XAI tools to elucidate decisions related to pediatric metabolic disease risk and heart disease classification, respectively. This emphasis on transparency is in alignment with the broader clinical imperative for interpretable systems that clinicians can integrate with confidence into diagnostic and management workflows.

3.4.2. Accountability and human oversight

A substantial number of studies have clearly delineated the role of AI as an adjunct to, rather than a replacement for human clinicians, thereby ensuring that ultimate decisions remain within the purview of professional supervision. Goh et al. (2024) and Pham et al. (2024) emphasize the necessity for human experts to interpret and validate model outputs in diagnostic reasoning tasks. Conversely, studies such as those by Gao et al. (2023) and Liu et al. (2024) have sought to anchor accountability in expert consensus by aligning AI predictions with recognized guidelines or biomedical literature. By requiring clinicians to ratify AI-generated recommendations, these articles reinforce the idea that accountability is shared between medical staff and the algorithms they deploy.

Table 4. Overview of which principles each article addresses from both the responsible/Trustworthy AI frameworks.

	Responsible AI (Arrieta et al., [18])						Trustworthy AI (EU Commission [20])						
Article	Fairness	Privacy	Accoutability	Ethics	Transparency	Security/safety	Human–agency	Tech–robust	Privacy/Data Governance	Transparency	Fairness	Societal well- being	Accoutability
Ahmed et al., 2024)	–	–	–	–	✓	✓	–	✓	–	–	–	–	–
Rao et al. (2023)	–	–	–	–	✓	–	✓	✓	–	✓	–	–	✓
Wada et al. (2024)	–	–	–	–	✓	–	✓	✓	–	✓	–	–	–
Lalk et al. (2024)	–	✓	–	–	✓	–	–	–	✓	✓	–	–	–
Arnaud et al. (2023)	–	✓	–	–	✓	–	✓	–	✓	✓	–	–	–
Dal et al. (2024)	–	–	–	–	✓	–	✓	–	–	✓	–	–	–
Goh et al. (2024)	–	–	–	–	–	–	✓	–	–	✓	–	–	✓
Hassan et al. (2024)	–	–	–	–	–	✓	–	✓	–	–	–	–	–
Diniz et al. (2022)	–	✓	✓	✓	–	–	✓	–	–	–	–	✓	–
Alloatti et al. (2021)	✓	✓	✓	✓	✓	✓	✓	–	–	✓	✓	✓	✓
Nazary et al. (2024)	–	–	–	✓	✓	✓	–	✓	–	✓	–	–	–
Shah-Mohammadi and Finkelstein (2024)	–	–	✓	✓	✓	✓	✓	✓	–	✓	–	–	–
Naseem and Catal (2024)	–	–	–	–	–	–	–	✓	–	✓	–	✓	–
Yang et al. (2024)	✓	–	–	✓	–	–	✓	✓	–	✓	✓	–	–
Zhang and Ogasawara (2023)	–	–	–	✓	✓	–	✓	–	–	✓	–	–	–
Chen et al. (2024)	–	–	–	–	✓	–	✓	✓	–	–	–	✓	–
Gao et al. (2023)	–	–	–	–	✓	–	✓	–	–	✓	–	✓	–
Pham et al. (2024)	✓	–	–	–	✓	–	✓	–	–	✓	–	✓	–
Li et al. (2024)	–	–	–	–	✓	–	✓	–	–	✓	–	✓	✓
Md Hasib et al. (2024)	–	–	–	–	✓	–	✓	–	–	✓	–	–	–
Rajashekar et al. (2024)	–	–	–	–	✓	–	✓	✓	–	–	–	✓	–
Oh et al. (2024)	–	–	–	–	✓	–	✓	–	–	✓	–	–	–
Chung et al. (2023)	–	–	–	–	✓	–	✓	–	–	✓	–	✓	–
Sun et al. (2024)	–	–	–	–	✓	–	✓	–	–	✓	–	✓	✓
Yang et al. (2023)	–	✓	–	–	✓	✓	✓	–	–	✓	–	–	✓
Griewing et al. (2024)	–	✓	–	–	✓	–	✓	–	–	✓	–	–	✓
Griewing et al. (2024)	–	–	–	–	✓	–	–	–	–	✓	–	–	–
Puts et al. (2023)	–	–	–	–	✓	–	–	–	–	✓	–	–	–
Rau et al. (2024)	–	–	–	–	✓	–	–	–	–	✓	–	–	–
Zhang and Song (2024)	–	–	–	–	✓	–	✓	–	–	✓	–	–	–
Savage et al. (2024)	–	–	–	–	✓	–	✓	–	–	✓	–	–	–
Fei et al. (2024)	–	–	–	–	✓	–	✓	–	–	✓	–	–	–
Meng et al. (2021)	–	–	–	–	✓	–	✓	✓	–	–	–	–	–
Rao et al. (2023)	–	–	–	–	✓	–	✓	✓	–	✓	–	–	–
Liu et al. (2024)	–	–	–	–	✓	–	✓	✓	–	✓	–	–	–
Wang et al. (2024)	–	–	–	–	✓	–	✓	✓	–	✓	–	–	–
Number of Studies Addressing each Principle	3	6	3	6	31	6	29	14	2	30	2	10	7

3.4.3. Ethics and patient-centered values

Ethical considerations frequently arise in the field of healthcare, reflecting the high-stakes nature of the sector and the moral responsibility to benefit patients. Research in this field has typically concentrated on the development of AI tools that are designed to reduce diagnostic errors, facilitate early interventions, and enhance clinical outcomes. For instance, Shah-Mohammadi and Finkelstein (2024) have developed an AI system that generates ED differential diagnoses that correspond with both hospital discharge outcomes and established medical standards, thus upholding ethical principles of beneficence and non-maleficence. In a similar vein, Diniz et al. (2022) emphasizes the importance of providing assistance to vulnerable through interpretable, clinically relevant models. In this manner, ethical objectives such as patient welfare, risk mitigation, and equitable service access serve as guiding principles in the implementation of AI in critical settings.

3.4.4. Privacy, fairness and security

Whilst transparency and accountability are emphasized to the greatest extent, a few number of articles address privacy, fairness and security directly. Lalk et al. (2024) and Alloatti et al. (2021), for instance, employ anonymized or secure methods to protect sensitive mental health data and diabetes management information, respectively. Fairness is addressed in works such as Yang et al. (2024), which examines demographic performance variations. Security-related strategies also emerge, for instance, Griewing et al. (2024) leverage local model hosting to maintain data privacy and foster a safe environment for AI deployment.

3.5. Trustworthy AI

The collective findings of the studies demonstrate that the development of Trustworthy AI in healthcare must attain a balance between the need for human control, interpretability, reliability, and ethical and equitable considerations (see Table 4). The most addressed principles were Transparency (n = 30 studies) and Human-Agency (n = 29 studies), and the least addressed were Fairness (n = 2 studies) and privacy/data governance (n = 2 studies). The necessity for transparency and human oversight is repeatedly emphasized, thereby ensuring that clinicians retain their central role in the decision-making process. The technical robustness and safety of the system are addressed through the implementation of rigorous performance evaluations and the incorporation of error mitigation strategies. Anonymization and secure data handling practices are employed in order to ensure compliance with privacy and data governance principles. While not all articles explicitly consider fairness, some incorporate fairness optimization techniques or design AI systems adaptable to diverse environments, thereby indirectly promoting equity. The improvement of healthcare efficiency, reduction of clinician workload and enhancement of patient outcomes are addressed through efforts to promote societal and environmental well-being. The accountability of AI models is upheld through benchmarking against expert standards and clinical guidelines, thereby reinforcing the idea that responsibility for patient care decisions ultimately resides with healthcare professionals.

3.5.1. Human oversight and accountability

A recurrent theme pervading the extant literature on the subject is that AI tools serve in an adjunct capacity, providing support to human clinicians rather than replacing them. Goh et al. (2024) and Li et al. (2024) emphasize that healthcare professionals must interpret and validate model outputs to ensure safe, context-appropriate decisions. In a similar vein, Yang et al. (2024) underscores the importance of human review, thereby maintaining the central role of clinical judgment. This human-in-the-loop arrangement is pivotal in safeguarding patient welfare and fostering trust, particularly given that ultimate responsibility remains with clinicians and healthcare organizations. The formalization of accountability is often achieved through the benchmarking of models against expert standards and recognized guidelines, as evidenced by Griewing et al. (2024), which aligns recommendations with clinical protocols, and Yang et al. (2023), which grounds AI suggestions in evidence-based literature.

3.5.2. Technical robustness and data governance

In clinical environments, where errors can result in significant harm, robustness and safety are paramount. The prevailing approach in such studies is to evaluate performance using metrics such as accuracy, AUC-ROC, cross-validation, or the judicious design of prompts to minimize errors. For instance, Ahmed et al. (2024) emphasized the efficacy of robust prompts in enhancing diagnostic reliability, while Rao et al. (2023) and Rau et al. (2024) evaluated GPT-based models against established clinical guidelines to ensure high accuracy. Adopting these practices enables AI systems to provide consistent and reliable outputs, thus enabling clinicians to rely on them with confidence in real-world settings.

Two research initiatives employ anonymized or de-identified data, thereby ensuring the confidentiality of patients. For instance, Lalk et al. (2024) utilize anonymized psychotherapy transcripts to safeguard sensitive mental health information. In addition to these approaches, further methods have been developed to reinforce data protection and governance, for instance, Griewing et al. (2024) host models locally to limit external access.

3.5.3. Transparency

Transparency is important in ensuring the reliability of AI, as it facilitates the comprehension of model logic and outcomes by clinicians. Techniques such as SHAP, LIME, Grad-CAM, and chain-of-thought prompting elucidate the manner in which predictions arise. Nazary et al. (2024) illustrate how such methods clarify decisions in heart disease classification or pediatric metabolic risk, respectively, whereas Savage et al. (2024) employ diagnostic reasoning prompts to enhance interpretability in large language models. This enhanced clarity fosters practitioner confidence, thereby paving the way for the responsible adoption of AI in daily clinical practice.

3.5.4. Fairness and societal well-being

Whilst not invariably a primary focus, a number of studies evaluate fairness by assessing model performance across different demographic groups or by employing fairness optimization. For instance, Yang et al. (2024) examine the extent to which AI generalizes to various populations, and Rajashekar et al. (2024) address global diversity by designing systems adaptable to both resource-rich and resource-scarce contexts. Other studies emphasize broader societal benefits. For instance, Diniz et al. (2022) support mental health professionals in preventing suicides, illustrating tangible societal impact, while Pham et al. (2024) underscore workflow efficiencies that improve patient outcomes and optimize healthcare resources. Although environmental sustainability is less frequently discussed, improved efficiency may indirectly reduce the carbon footprint of healthcare operations.

4. Discussion

4.1. RQ1. What are the current main solutions on CDSSs with AI?

The current landscape of AI-driven CDSSs encompasses different approaches, each offering distinct contributions to medical decision-making. Generative LLMs, such as GPT-4, together with transformer-based encoder models (e.g. BioBERT, ClinicalBERT, MCN-BERT, and domain-specific adaptations such as FlauBERT and CamemBERT, have become prevalent in clinical settings (see Figure 3). To contextualize recent developments beyond the temporal scope of the systematic search, we additionally reference selected emerging studies published after the review cutoff date. These works are discussed as related developments illustrating the rapid evolution of generative LLM-based CDSSs, but were not part of the systematically selected dataset analyzed in the methodology section. These models have been employed to enhance medical reasoning, facilitate diagnosis, automate the extraction of medical information from electronic health records, and improve the interpretation of medical reports. Beyond reasoning tasks, LLMs have also been applied to the structuring and harmonization of unstructured clinical documentation.

For example, automated pipelines leveraging GPT-based architectures have demonstrated the capacity to extract and standardize neuropathological findings from heterogeneous reports, thereby supporting downstream analytics and improving data interoperability in neurological research (Stroganov

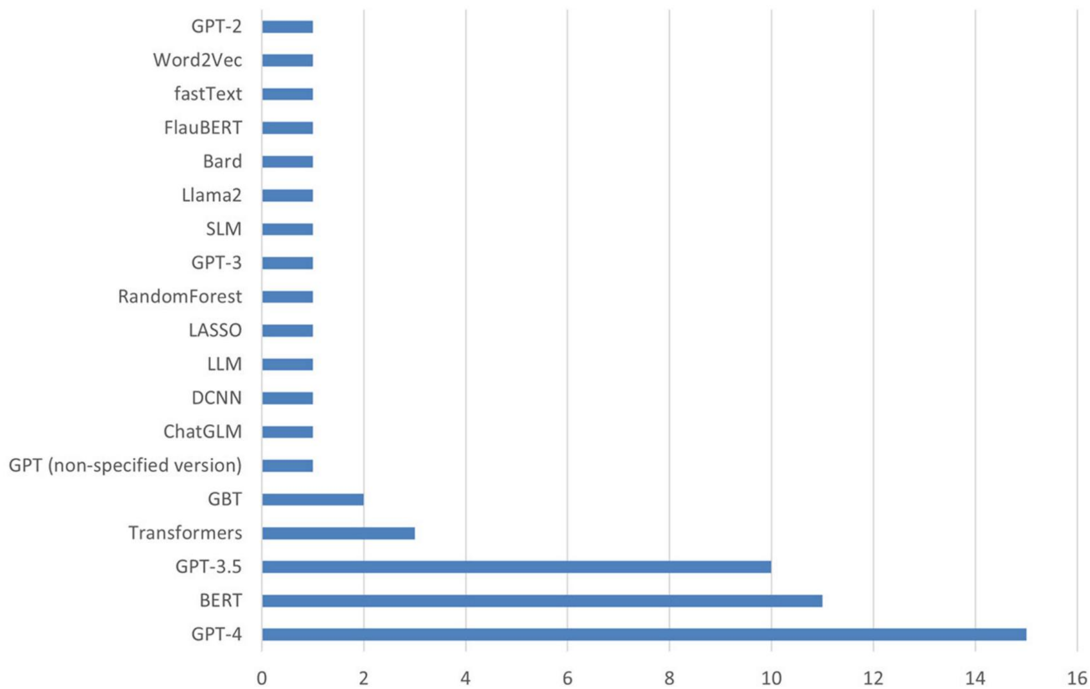

Figure 3. Bar chart illustrates the primary technologies identified in the selected studies, along with the count of studies employing each technology. Note that individual studies may span multiple technologies, so totals can exceed the number of studies.

et al., 2024). Their applications span a wide range, including medical triage, where GPT-3.5 and GPT-4 have demonstrated effectiveness in guiding emergency decision-making (Rao et al., 2023), to predicting hospital readmissions using BioBERT (Naseem & Catal, 2024) and assisting in medical coding by mapping clinical concepts to ICD-10 with ClinicalBERT (Gao et al., 2023).

Concurrently, conventional machine learning techniques, encompassing RF, XGBoost, Logistic Regression, and Support Vector Machines, persist in prevalent utilization within the domain of structured clinical data analysis, a consequence of their computational efficiency, interpretability, and robustness. These models have been successfully applied in diverse scenarios, including predicting whether a patient requires further testing in the emergency department (Shahbandegan et al., 2022), assessing infection risk in pediatric leukemia patients (Al-Hussaini et al., 2024), and automating the classification of preoperative patient status using ASA-PS scores (Chung et al., 2023).

Deep learning models, particularly CNNs and Recurrent Neural Networks, such as long short-term memory (LSTM) and gated recurrent units (GRUs), have been extensively employed for processing complex clinical data, including medical images, time-series physiological signals, and unstructured clinical text. These models have demonstrated a high degree of efficacy in imaging-based diagnostics, such as pathology and radiology analysis. LSTMs and GRUs are widely used for analyzing sequential clinical data, including cardiac anomaly detection (Md Hasib et al., 2024) and the automated interpretation of clinical narratives through BiLSTM and CNN-based embeddings (Leroy et al., 2024). These deep learning methods frequently incorporate explainability techniques, such as Grad-CAM and SHAP, to enhance transparency by visualizing key decision-making features. Concurrently, transformer-based models, specifically trained on clinical datasets, including ClinicalBERT, BioBERT, and MCN-BERT, are employed to enhance predictive accuracy in various healthcare applications. It is important to distinguish between transformer-based encoder models (e.g. BioBERT, ClinicalBERT), which are primarily optimized for classification and information extraction tasks, and generative large language models capable of multi-step reasoning, contextual summarization, and interactive response generation. For instance, a clinical decision support system for syncope recognition in emergency settings employed domain-adapted BERT architectures to perform structured classification from electronic medical records with high discriminative performance (Levra et al., 2025). While such approaches demonstrate strong predictive capabilities within defined workflows, they differ conceptually from generative LLM-based CDSSs that support dynamic reasoning and therapeutic planning. These domain-adapted models have been effectively utilized for tasks such as hospital admission prediction (Naseem & Catal, 2024), clinical concept normalization for standardized medical records (Hassan et al., 2024), and the generation of explainable clinical reports to support physician decision-making (Lalk et al., 2024). Despite the advancements in LLMs and deep learning, classical NLP techniques remain a fundamental component of AI-driven CDSSs. Approaches such as TF-IDF, Word2Vec, and FastText persist in their utilization for tasks involving feature extraction and text classification, particularly within the context of structured clinical documentation and linguistic analysis. Another critical aspect of AI-enabled CDSSs is the implementation of XAI techniques to enhance transparency and user trust. SHAP and LIME are extensively utilized in machine learning and deep learning models to assist clinicians in comprehending the rationale behind AI-generated diagnoses, thereby enhancing interpretability and facilitating informed decision-making (Arnaud et al., 2023). In the domain of medical imaging, Grad-CAM is frequently employed to accentuate the regions of interest that influenced diagnostic predictions, thereby enhancing clinicians' capacity to validate AI-driven assessments (Abu Sufian et al., 2024). In a similar vein, Chain-of-Thought prompting has been integrated into LLM-based models to align their responses more closely with human clinical reasoning, fostering greater interpretability and consistency in decision support (Savage et al., 2024). More recent investigations indicate that generative LLMs are progressively transitioning from experimental evaluation to clinically contextualized deployment. Prospective comparative analyses in emergency internal medicine have shown that advanced reasoning models, including OpenAI's o1, Claude 3.5, and Llama 3.2, can achieve diagnostic and therapeutic performance levels comparable to human specialists (Vrdoljak et al., 2025). Similarly, evaluations in clinicopathological conferences have demonstrated that GPT-4 and Google Bard are capable of generating structured differential diagnoses aligned with expert reasoning processes (Koga et al., 2024). Beyond diagnostic assistance, generative frameworks integrating Retrieval-Augmented Generation (RAG) and structured reasoning strategies have been applied to personalized Parkinson's disease treatment optimization, achieving measurable improvements in clinical outcome indicators (Zhang et al., 2026).

Furthermore, knowledge-based decision support systems, which integrate structured medical knowledge bases, clinical guidelines, and biomedical ontologies, play a crucial role in ensuring the reliability of AI-generated recommendations. These systems enhance AI-assisted decision-making by aligning predictive models with established medical protocols, reducing potential errors, and improving trust in automated recommendations. Examples of this integration include the use of oncology treatment guidelines within artificial intelligence models for the assessment of breast cancer risk and treatment planning (Griewing et al., 2024); the development of hybrid systems that combine machine learning predictions with clinical rules for the evaluation of thyroid nodules in ThyroAIGuide (Wang et al., 2024); and the use of evidence-based clinical chatbots designed to provide guideline-driven responses for patient consultations (Rau et al., 2024). Collectively, these technological advancements demonstrate that the most effective AI-driven CDSSs not only leverage state-of-the-art machine learning and NLP techniques but also adhere to some of the responsible/trustworthy AI principles—including transparency, security, ethics, and interpretability. In parallel, large-scale hospital implementations have demonstrated the feasibility of embedding frontier LLMs directly within Electronic Health Record systems under strict governance safeguards. A recent deployment integrating Qwen3-235B with Retrieval-Augmented Generation into an Epic EHR environment achieved sustained clinician adoption while operating within a GDPR-compliant, on-premises infrastructure (Griot et al., 2025). Such implementations illustrate the convergence of generative AI capabilities with secure ETL pipelines, structured data retrieval, and responsible AI principles in routine clinical workflows.

4.2. RQ2. What research topics are being addressed in the intersection between generative AI, responsible AI/Trustworthy AI and CDSSs?

The articles collectively illustrate the complexity of implementing Responsible AI in healthcare. Transparency and accountability are frequently prioritized to ensure that AI complements physician judgment (Antoniadi et al., 2021; Barredo Arrieta et al., 2020). Ethical considerations and privacy compliance are reflected in the sensitive nature of patient data and the moral responsibility of improving health outcomes (Veale & Zuiderveen Borgesius, 2021). Fairness and security, while sometimes less emphasized, are beginning to receive attention as developers recognize the importance of equitable, secure AI tools (Yang et al., 2024). A collective analysis of the extant literature reveals a discernible shift toward the adoption of responsible AI in healthcare, with transparency and interpretability emerging as predominant themes (Nazary et al., 2024). This ensures that clinicians can comprehend and trust AI-driven recommendations. The accountability of AI is maintained by framing it as a support tool that is subject to human oversight (Goh et al., 2024; Pham et al., 2024). The research objectives are informed by an ethical framework that prioritizes improvements in patient care, privacy, and equitable access (Diniz et al., 2022). While fairness and security are not universally addressed, some studies integrate these principles (Griewing et al., 2024).

As the field of healthcare AI continues to evolve, it is reasonable to anticipate that responsible AI principles will become increasingly comprehensive. This progression is expected to guarantee that healthcare AI not only demonstrates technical excellence but also adheres to ethical, transparent, and patient-centric standards. A comprehensive review of the extant literature reveals a multifaceted engagement with a plethora of principles pertaining to Trustworthy AI. The articles under review demonstrate a clear emphasis on transparency, human oversight, and technical robustness, which reflect the priority of interpretability, reliability, and human control in healthcare (Hassan et al., 2024; Savage et al., 2024). The implementation of secure data practices is crucial for upholding the principles of data privacy and governance (Lalk et al., 2024). An evaluation was conducted to ascertain the fairness of the proposed AI solution in psychiatric rehabilitation contexts (Yang et al., 2024). This evaluation considered potential biases in processing data from various demographic groups (e.g. race, age and sex). Furthermore, the concept of accountability emerges as a pivotal factor, with healthcare professionals ultimately evaluating and trusting AI recommendations (Griewing et al., 2024; Yang et al., 2023). The integration of AI into healthcare workflows holds the potential to enhance societal well-being by facilitating the delivery of timely, effective, and equitable care to patients (Diniz et al., 2022; Pham et al., 2024). As healthcare AI solutions continue to evolve, these Trustworthy AI principles will serve as a framework for their responsible development and deployment.

4.3. RQ3. What are the limitations of current research, and how should we approach them?

Notwithstanding the substantial advancements in AI-driven CDSSs, numerous pivotal limitations persist in impeding their extensive implementation within medical practice. A salient concern pertains to the generalization capability of AI models, generative LLMs (e.g. GPT-4) and transformer-based encoder models (e.g. BioBERT, ClinicalBERT). These models are predominantly trained on region-specific datasets that may not adequately represent the heterogeneity of patient populations. This can result in biased model outputs, lower accuracy in non-Western clinical environments, and reduced adaptability to different healthcare systems (Arnaud et al., 2023; Digumarthi et al., 2024). Moreover, the opaque nature of deep learning models and transformers persists as a salient concern, as the absence of transparency in decision-making processes can impede clinical trust. While XAI techniques, including SHAP, LIME, and Grad-CAM, have been employed to enhance interpretability, they are inadequate in fully elucidating the process by which AI-driven recommendations are generated (Abu Sufian et al., 2024; Javidi et al., 2024). Clinicians often require clear justifications for AI-based decisions before integrating them into practice, yet current models struggle to provide reliable explanations (Pham et al., 2024; Wilcox et al., 2023).

A significant challenge pertains to the integration of AI models with existing clinical systems, such as electronic health records. A significant number of AI solutions are developed in isolation, which poses significant interoperability challenges that complicate their deployment within hospital workflows (Chen et al., 2024; Leroy et al., 2024). Moreover, concerns regarding ethics, regulation, and privacy persist, as the extensive utilization of patient data necessitates adherence to stringent data protection regulations such as the General Data Protection Regulation (Ryz & Grest, 2016) or the Health Insurance Portability and Accountability Act (Wilcox et al., 2023). Additionally, the training of AI models on non-representative datasets can lead to the amplification of biases and the generation of discriminatory outcomes, which can have a disproportionate impact on vulnerable patient populations (Digumarthi et al., 2024; Yang et al., 2024). Finally, a substantial gap in clinical validation persists, as many AI-based CDSSs have yet to undergo rigorous real-world testing. While certain models exhibit strong performance in controlled environments, their applicability to diverse clinical settings remains uncertain due to a paucity of multicenter studies and long-term evaluations (Fei et al., 2024; Rao et al., 2023). The distribution of evidence levels indicates that the majority of studies remain at an intermediate stage of development. While many studies leverage real clinical data, most are limited to retrospective validation without integration into clinical workflows. Only a smaller subset reaches higher levels of maturity, involving clinician interaction or evaluation in realistic clinical contexts. This highlights a gap between technical performance and real-world deployment, suggesting that further work is needed to translate AI-driven CDSS from experimental settings into routine clinical practice. Addressing these limitations will require greater emphasis on model transparency, standardized evaluation frameworks, regulatory clarity, and seamless integration with clinical workflows to ensure that AI-driven CDSSs can be reliably and effectively implemented in healthcare settings.

In order to address the limitations in current research, it is crucial to advance the design and evaluation of AI-driven CDSSs through a human-centered approach. This process entails the active involvement of clinicians and end-users throughout all phases of system development, encompassing the initial requirements gathering, iterative prototyping, and deployment stages. Adopting participatory and co-design methodologies is a recommended approach for developers seeking to ensure that AI recommendations are not only technically accurate but also interpretable, relevant, and actionable within real clinical workflows. The design of explanation interfaces (based on XAI methods for the explainability of the LLMs) and feedback mechanisms that align with clinicians' decision-making processes has the potential to further enhance trust, transparency, and safe adoption in practice. For example, one study examined the potential of interactive explanation interfaces to foster trust and transparency by enabling clinicians to query or challenge model outputs (Yang et al., 2024). These practices facilitate not only technical interpretability, but also human oversight and practical adoption and integration in everyday workflows, ultimately reinforcing the human-centered nature of HCAI. Furthermore, the promotion of data representativeness and fairness necessitates the incorporation of diverse patient populations during model training and ongoing monitoring for bias in real-world settings. It is imperative that seamless integration with electronic health records and adherence to interoperability standards are prioritized in order to minimize workflow disruptions. Finally, robust governance frameworks – encompassing ethical oversight, regulatory compliance, and continuous post-deployment evaluation – are essential to ensure that AI-based CDSSs remain aligned with responsible and trustworthy AI principles.

5. Conclusions, limitations and future work

This systematic literature review examined the incorporation of generative AI, particularly LLMs, into CDSSs. The review evaluated the alignment of these systems with Responsible and Trustworthy AI principles. Employing a structured methodology, we identified 36 relevant studies. The findings reveal that AI-driven CDSSs leverage various computational methods, including traditional machine learning, deep learning, NLP, and knowledge-based decision support, alongside specialized techniques such as XAI. While these approaches show potential for improving decision-making in clinical practice, challenges persist around model generalization, interpretability, integration with clinical workflows, and ethical and regulatory compliance. Additionally, there is an increasing emphasis on ensuring that AI models are transparent, fair, and accountable—qualities that are critical in high-stakes healthcare contexts. Despite promising results, this review highlights significant barriers to the widespread adoption of AI-based CDSSs. The challenges include issues related to generalizability and clinical validation, as many studies are constrained by domain-specific datasets prone to bias, thereby limiting applicability across diverse patient populations. Another significant challenge pertains to interpretability, which poses a particular obstacle in the context of deep learning and LLM-based models. Even advanced XAI tools (e.g. SHAP, LIME, Grad-CAM) frequently fall short of achieving clinical trust. From an implementation perspective, healthcare institutions face hurdles in system interoperability, workflow integration, and usability, all of which can undermine real-world utility. Concurrently, ethical and regulatory constraints, including data privacy, security, and bias mitigation, necessitate the establishment of robust governance frameworks, such as the GDPR or the HIPAA. Presently, the majority of models remain in the proof-of-concept stage, necessitating extensive clinical testing before their confident adoption in practice.

This review is subject to several limitations. Firstly, the inclusion criteria may have been constrained by variations in terminology and indexing practices, potentially excluding relevant studies that do not explicitly identify the use of generative AI or Responsible AI principles. Secondly, incomplete or inconsistent reporting in some articles—particularly concerning evaluation methods—limited a full assessment of how well AI models perform in clinical settings. Thirdly, although ethical considerations were a fundamental component of this review, many studies offered only fragmentary insights on fairness and equity, complicating efforts to determine AI's broader impact on healthcare disparities.

Future research work could aim to prioritize transparency and interpretability in AI models, thereby enabling clinicians to comprehend and repose their trust in algorithmic outputs. The field would benefit from the implementation of standardized evaluation frameworks, which would foster broader generalization and simplify cross-institutional adaptation. Regulatory guidelines concerning data security, bias prevention, and fairness also require continued refinement to promote responsible AI deployment at scale. Beyond technical improvements, further studies should explore multimodal AI methods that integrate both structured and unstructured clinical data, accompanied by longitudinal, multi-institutional trials to validate efficacy in real-world environments. Addressing these challenges can pave the way for safe, effective, and responsible AI-based CDSSs, ultimately enhancing clinical decision-making and patient outcomes.

Notes

2. Proposal for a Regulation laying down harmonised rules on artificial intelligence: Shaping Europe's digital future. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A52021PC0206.
3. https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf.

Authors' contributions

CRediT: **Dennis Paulino**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **André T.C. Netto**: Data curation, Formal analysis, Investigation, Methodology, Resources; **Rafael Ris-Ala**: Formal analysis, Investigation, Validation, Writing – review & editing; **Artur Rocha**: Funding acquisition, Project administration; **Hugo Paredes**: Conceptualization, Funding acquisition, Methodology, Project administration, Validation.

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ORCID

Dennis Paulino http://orcid.org/0000-0001-8463-6302
André T.C. Netto http://orcid.org/0009-0001-5003-7421
Rafael Ris-Ala http://orcid.org/0000-0003-3182-7724
Artur Rocha http://orcid.org/0000-0002-5637-1041
Hugo Paredes http://orcid.org/0000-0002-4274-4783

Data availability statement

The data supporting the findings of this study, including screening records and supplementary extraction materials, are publicly available on Zenodo at: https://doi.org/10.5281/zenodo.20599096

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About the authors

Dennis Paulino obtained his PhD in Informatics at UTAD. He currently serves as an Auxiliar Researcher at INESC TEC, where his research interests in HCI, particularly in Accessibility and Digital Work. He has authored over 40 publications in international journals and conferences and participated in numerous international R&D projects.

André Thiago Netto is Research Assistant at INESC TEC and a PhD candidate in Computer Science at UTAD. His research focuses on Human–Computer Interaction, digital phenotyping, and artificial intelligence for mental health. His work explores behavioral biomarkers and human-centered technologies to support early screening, assessment, and personalized care.

Rafael Ris-Ala is a researcher at INESC TEC and a doctoral candidate at UFRJ under a cotutelle agreement with UTAD. His research focuses on Software Engineering, Artificial Intelligence, and Trustworthy Large Language Models (LLMs). He is a Professor at USP and PUC Minas and an author published by Springer Nature.

Artur Rocha is a Coordinating Researcher at INESC TEC and co-coordinator of Humanize. He holds a PhD in Informatics Engineering and has led research on collaborative digital platforms, privacy-preserving computing, and data-intensive infrastructures. His work advances human-centered and trustworthy digital solutions for personalized health and scientific research.

Hugo Paredes is a Full Professor at University of Tras-os-Montes e Alto Douro (UTAD) and coordinator of Center for Human-Centered Computing and Information Science (HumaISE) at INESC TEC. His main research interests are in the domain of Human-Centered AI, with application to climate change, health, and active and healthy aging.